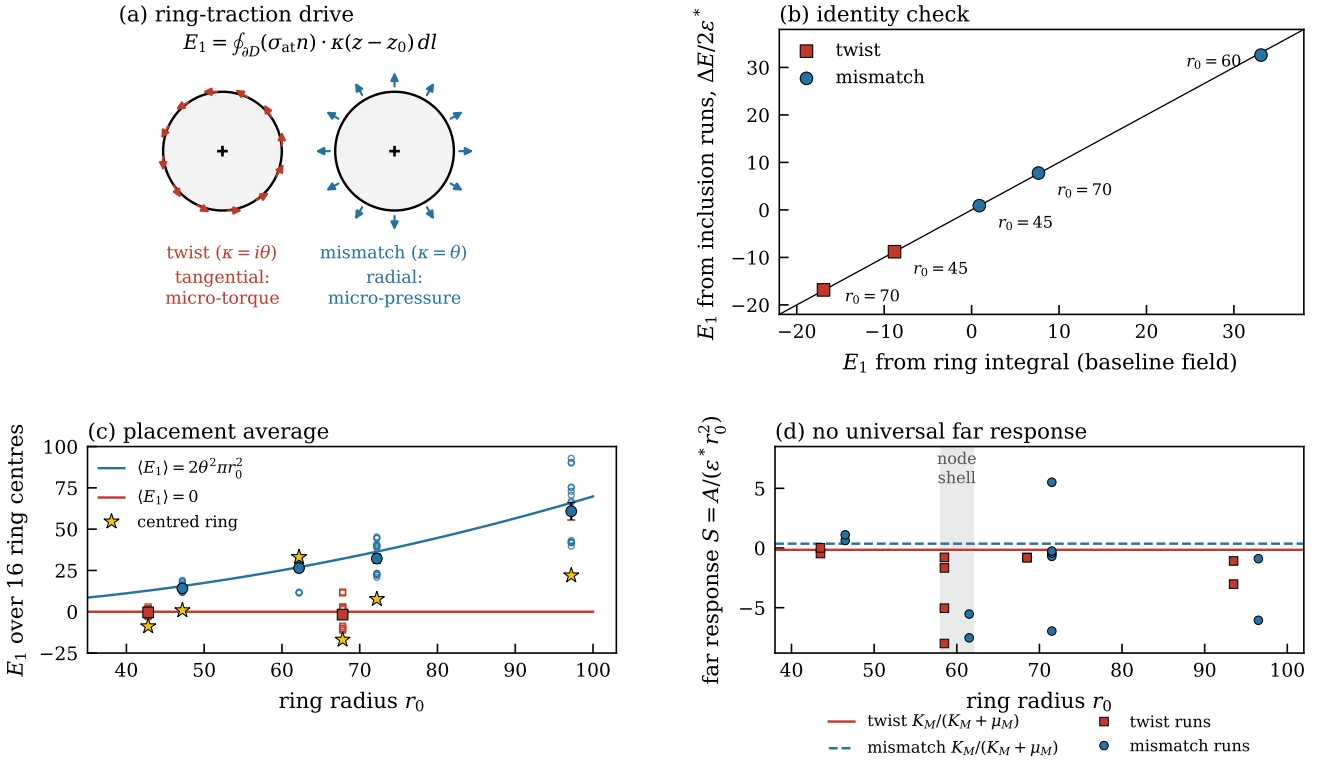
(a) ring-traction drive
$E_1 = \oint_{\partial D} (\sigma_{at} n) \cdot \kappa (z - z_0)\, dl$
twist ($\kappa = i\theta$)
tangential:
micro-torque
mismatch ($\kappa = \theta$)
radial:
micro-pressure
(b) identity check
twist
mismatch
$r_0 = 60$
$r_0 = 70$
$r_0 = 45$
$r_0 = 45$
$r_0 = 70$
E_1 from inclusion runs, $\Delta E / 2\varepsilon^*$
E_1 from ring integral (baseline field)
(c) placement average
$\langle E_1 \rangle = 2\theta^2 \pi r_0^2$
$\langle E_1 \rangle = 0$
centred ring
E_1 over 16 ring centres
ring radius r_0
(d) no universal far response
node
shell
far response $S = A/(\varepsilon^* r_0^2)$
ring radius r_0
twist $K_M/(K_M + \mu_M)$
mismatch $K_M/(K_M + \mu_M)$
twist runs
mismatch runs